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Inventor(s)	Gunasekaran; Mohan et al.

DOWNHOLE TOOL INCLUDING A FREQUENCY FILTER BASED SWITCH SYSTEM CONFIGURED TO FILTER POWER BETWEEN A FIRST DOWNHOLE DEVICE AND A SECOND DOWNHOLE DEVICE

Abstract

Provided is a downhole tool, a well system, and a method. The downhole tool, in one aspect, includes a first downhole device. The downhole tool, in one aspect, further includes a switch system electrically coupled with the first downhole tool, the switch system including: 1) an input coupleable to a power source; 2) an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and 3) a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source.

Inventors: **Gunasekaran; Mohan (Al-Khobar, SA), Fripp; Michael Linley (Singapore, SG), Upadrashta; Deepesh (Singapore, SG), Dockweiler; David (Singapore, SG), El Mallawany; Ibrahim (Dhahran, SA)**

Applicant: **Halliburton Energy Services, Inc.** (Houston, TX)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application Ser. No. 63/559,011, filed on Feb. 28, 2024, entitled “A WIRELINE RETRIEVABLE SAFETY VALVE EMPLOYING A MECHANICAL CONNECTING APPARATUS HAVING ONE OR MORE PERMANENT MAGNETS,” U.S. Provisional Application Ser. No. 63/559,047, filed on Feb. 28, 2024, entitled “A WIRELINE RETRIEVABLE SAFETY VALVE EMPLOYING A MAGNETIC FLUX AND FLUX PATH OF AN ELECTROMAGNET TO ENGAGE WITH A MECHANICAL CONNECTING APPARATUS HAVING A FERROMAGNETIC TARGET,” and U.S. Provisional Application Ser. No. 63/559,031, filed on Feb. 28, 2024, entitled “A WIRELINE RETRIEVABLE SAFETY VALVE EMPLOYING RADIALLY COUPLED PERMANENT MAGNETS AND AN ELECTROMAGNET AXIALLY COUPLED TO A TARGET,” all of which are commonly assigned with this application and incorporated herein by reference in their entirety.

BACKGROUND
[0002] Downhole devices, such as subsurface safety valves (SSSVs) are well known in the oil and gas industry and provide one of many failsafe mechanisms to prevent the uncontrolled release of subsurface production fluids, should a wellbore system experience a loss in containment. In certain instances, SSSVs comprise a portion of a tubing string, the entirety of the SSSVs being set in place during completion of a wellbore. In other instances, the SSSVs are wireline deployed/retrieved. Although a number of design variations are possible for SSSVs, the vast majority are flapper-type valves that open and close in response to longitudinal movement of a flow tube.
[0003] Since SSSVs typically provide a failsafe mechanism, the default positioning of the flapper valve is usually closed in order to minimize the potential for inadvertent release of subsurface production fluids. The flapper valve can be opened through various means of control from the earth's surface in order to provide a flow pathway for production to occur. What is needed in the art is an improved SSSV that does not encounter the problems of existing SSSVs.

Description

BRIEF DESCRIPTION
[0004] Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:
[0005] FIG. **1**A illustrates a well system designed, manufactured and/or operated according to one or more embodiments of the disclosure;

[0006] FIGS. **1**B and **1**C illustrate one embodiment of a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system of FIG. **1**A;
[0007] FIGS. **1**D and **1**E illustrate an alternative embodiment of a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system of FIG. **1**A;
[0008] FIGS. **1**F and **1**G illustrate an alternative embodiment of a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system of FIG. **1**A;
[0009] FIGS. **1**H and **1**I illustrate an alternative embodiment of a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system of FIG. **1**A;
[0010] FIGS. **1**J and **1**K illustrate an alternative embodiment of a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system of FIG. **1**A;
[0011] FIG. **1**L illustrates a chart showing various different ways that an operator may provide power to a TRSV and/or WLRSV, including using a single electric control line, two dedicated electric control lines, a single electric control line with a switch, as well as a single electric control line with low/high frequency filters;
[0012] FIGS. **2**A through **2**F illustrate one embodiment of a safety valve designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might employ first, second and third portions of a WLRSV;
[0013] FIGS. **3**A through **3**D illustrate different views of a safety valve designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure; and
[0014] FIGS. **4**A through **9**D illustrate various different installation states, each with various different views, of a safety valve designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure.

DETAILED DESCRIPTION

[0015] In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms. Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results.
[0016] Unless otherwise specified, use of the terms “connect,” “engage,” “couple,” “attach,” or any other like term describing an interaction between elements is not meant to limit the interaction to direct interaction between the elements and may also include indirect interaction between the elements described. Furthermore, unless otherwise specified, use of the terms “up,” “upper,” “upward,” “uphole,” “upstream,” or other like terms shall be construed as generally toward the surface of the subterranean formation; likewise, use of the terms “down,” “lower,” “downward,” “downhole,” “downstream,” or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical axis. Additionally, unless otherwise specified, use of the term “subterranean formation” shall be construed as encompassing both areas below exposed earth and areas below earth covered by water such as ocean or fresh water.

[0017] Various values and/or ranges are explicitly disclosed in certain embodiments herein. However, values/ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited. Similarly, values/ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited. In the same way, values/ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values even if not explicitly recited. Thus, every point or individual value may serve as its own lower or upper limit combined with any other point or individual value or any other lower or upper limit, to recite a range not explicitly recited. Similarly, an individual value disclosed herein may be combined with another individual value or range disclosed herein to form another range.
[0018] The term “substantially XYZ,” as used herein, means that it is within 10 percent of perfectly XYZ. The term “significantly XYZ,” as used herein, means that it is within 5 percent of perfectly XYZ. The term “ideally XYZ,” as used herein, means that it is within 1 percent of perfectly XYZ. The monicker “XYZ” could refer to parallel, perpendicular, alignment, or other relative features disclosed herein.
[0019] The present disclosure has acknowledged that offshore wells are being drilled at ever increasing water depths and in environmentally sensitive waters, and thus safety valves (e.g., including subsurface safety valves (SSSVs)) are necessary. The present disclosure has further acknowledged that SSSVs have parts that can wear or erode, and thus from time to time may need servicing and/or replacing. In fact, occasionally the tubing retrievable safety valve (TRSV) (e.g., electrically actuated TRSV) will fail, and then a wireline retrievable safety valve (WLRSV) will be run in hole. Unfortunately, each of the TRSV and the WLRSV require their own power source, such as individual tubing encapsulated conductors (TECs).
[0020] The present disclosure has further developed an improved WLRSV. In at least one embodiment, the WLRSV includes a first portion that is run-in-hole with the TRSV and second and third portions that are run-in-hole after the TRSV is no longer working properly and/or has failed. The first portion of the WLRSV, in at least one embodiment, includes a safety valve sub (e.g., WLRSV sub) that would be run-in-hole along with another safety valve sub (e.g., TRSV sub), and for example the tubing string. In at least one embodiment, the safety valve sub would be located above the TRSV sub. In at least one other embodiment, the safety valve sub would include an electromagnetic assembly (e.g., including one or more coils) (e.g., coupleable to the electric control line (e.g., single TEC) via the aforementioned switch system) located in an pocket of the safety valve sub, as well as one or more radial outer magnetic targets and a magnetic target additionally located in the pocket, the magnetic target positioned between the one or more radial outer magnetic targets and the electromagnetic assembly. In at least one embodiment, the first portion additionally includes a fluid isolation sleeve that isolates the electromagnetic assembly, one or more radial outer magnetic targets, magnetic target and pocket from fluid and/or debris within the wellbore. In at least one embodiment, the fluid isolation sleeve is a fixed fluid isolation sleeve, and thus does not readily move once positioned downhole.
[0021] The WLRSV, in one or more embodiments, further includes the second portion of the WLRSV, which is run-in-hole after the TRSV is no longer working properly and/or has failed. The second portion of the WLRSV, in accordance with one or more embodiments, may be run-in-hole within the TRSV, for example using a latch mechanism to axially fix the second portion of the WLRSV within the TRSV. The second portion of the WLRSV, in one or more embodiments, may include a bore flow management actuator and a valve closure mechanism, and may be located below the first portion of the WLRSV including the electromagnetic assembly and the fluid

isolation sleeve. The second portion of the WLRSV may additionally include a power spring and/or nose spring, as will be further discussed below

[0022] The WLRSV, in one or more embodiments, further includes a third portion that is run-in-hole after the second portion of the WLRSV is latched downhole (e.g., latched within the TRSV or first portion of the WLRSV). In another embodiment, the third portion is run-in-hole after the second portion is run-in-hole on a separate wellbore operation, such as a separate wireline trip or a separate slickline trip. In another embodiment, the third portion is run-in-hole after the second portion in the same wellbore operation, such as on the same wireline trip or the same slickline trip. The third portion, in one or more embodiments, includes a mechanical connecting apparatus. For example, in accordance with one or more embodiments of the disclosure, once the second portion of the WLRSV is latched in place, the mechanical connecting apparatus may be run-in-hole and coupled with the flow tube of the second portion. In at least this one embodiment, the mechanical connecting apparatus is located radially inside of the electromagnetic assembly and/or the fluid isolation sleeve of the first portion. The mechanical connecting apparatus, in one or more embodiments, includes one or more radial inner magnetic targets associated therewith (e.g., coupled thereto or forming a part thereof). The one or more radial inner magnetic targets, in this embodiment, are configured to magnetically couple with the one or more radial outer magnetic targets in the pocket of the first portion (e.g., through the fluid isolation sleeve). Accordingly, in at least one embodiment, at least one of the one or more radial inner magnetic targets or the one or more radial outer magnetic targets are one or more permanent magnets. For example, in at least one embodiment, the one or more radial inner magnetic targets are one or more permanent magnets and the one or more radial outer magnetic targets are not permanent magnets. In at least one other embodiment, the one or more radial outer magnetic targets are one or more permanent magnets, and the one or more radial inner magnetic targets are not one or more permanent magnets. In even yet another embodiment, the one or more radial inner magnetic targets and the one or more radial outer magnetic targets are both one or more permanent magnets. In essence, the mechanical connecting apparatus may be run-in-hole to axially fix the one or more radial inner magnetic targets of the third portion with the flow tube of the second portion. Accordingly, any axial movement of the flow tube would result in the same axial movement of the one or more radial inner magnetic targets, and vice-versa. Similarly, as the one or more radial inner magnetic targets arc coupled (e.g., magnetically coupled) with the one or more radial outer magnetic targets in the pocket of the first portion, the axial movement of the one or more radial inner magnetic targets (e.g., through the movement of the flow tube) also axially slides the one or more radial outer magnetic targets of the first portion (e.g., and the magnetic target) toward the electromagnetic assembly of the first portion. Ultimately, when the magnetic target and the electromagnetic assembly of the first portion are positioned proximate one another, and the electromagnetic assembly is energized, the magnetic target and the electromagnetic assembly of the first portion are held together, thereby holding the mechanical connecting apparatus and bore flow management actuator of the second portion in the open position as long as the electromagnetic assembly is energized. The above is discussed in the context of the second portion and the third portion being run-in-hole at different times. Other embodiments may exist wherein the second portion and the third portion are run-in-hole in a single trip (e.g., already coupled with one another).

[0023] In operation, once the mechanical connecting apparatus is in place, fluid pressure (e.g., from within the tubular below the valve closure mechanism) may urge the bore flow management actuator toward the valve closure mechanism. Typically, the bore flow management actuator is unable to move past the valve closure mechanism until a pressure differential across the valve closure mechanism is reduced/eliminated. Once the pressure differential across the valve closure mechanism is reduced/eliminated, for example by pumping fluid down the wellbore toward an uphole side of the valve closure mechanism, the bore flow management actuator may be urged past the valve closure mechanism, for example using one or more springs (e.g., power springs and/or nose springs). As the one or more radial inner magnetic targets are axially fixed to the flow tube, the axial movement of the flow tube also axially moves the one or more radial inner magnetic targets. As the one or more radial inner magnetic targets are coupled to the one or more radial outer magnetic targets (e.g., and the magnetic target) of the first portion, the axial movement of the flow tube also axially moves the one or more radial outer magnetic targets (e.g., and the magnetic target). Therefore, this axial movement of the flow tube brings the one or more radial outer magnetic targets and magnetic target of the first portion proximate the electromagnetic assembly of the first portion. Accordingly, when the electromagnetic assembly is energized (e.g., before, during or after the one or more radial outer magnetic targets approach the electromagnetic assembly) and located proximate the magnetic target of the first portion, the one or more radial outer magnetic targets of the sliding sleeve, one or more radial inner magnetic targets of the mechanical connecting apparatus, and thus the flow tube axially fixed thereto, may be held in the flow state. The one or more radial inner magnetic targets of the mechanical connecting apparatus and the associated flow tube will be held in this flow state until such time as the electromagnetic assembly is no longer energized, such as when power is turned off to or cut from the electromagnetic assembly. When this happens, the one or more springs (e.g., power springs and/or nose springs) are allowed to return the flow tube, and the associated flapper valve, to the closed state.

[0024] The present disclosure has, for the first time, further developed a switch system (e.g., mechanical, electrical, etc.) that will allow a single electric control line (e.g., single TEC) to operate two different downhole tools, such as the TRSV (e.g., electrically actuated TRSV) and/or WLRSV (e.g., a WLRSV that may be electrically maintained in an open position), or to operate redundant downhole tools, such as a wet connection or an actuator. For example, the switch system could shift power between two different electrical devices (e.g., electromagnetic coils, electric motor or pump, piezoelectric actuator, solenoid valve, etc.) of the two different downhole tools. As another example, the switch system could shift power between an electrical device that has failed to a redundant device that has not been powered. Thus, in at least one embodiment, the single electric control line (e.g., single TEC) could be run downhole from the surface to the switch system, and then the switch system would toggle the power between the TRSV and the WLRSV, as necessary. In at least one embodiment, the switch system would toggle the power from the TRSV to the WLRSV as the WLRSV is ready to be run-in-hole, as the WLRSV is being run-in-hole, or after the WLRSV has been run-in-hole.

[0025] Accordingly, a switch system designed, manufactured and/or operated according to one or more embodiments of the disclosure reduces the need to run additional electric control lines, for example in contingency operations, such as when the TRSV fails and a WLRSV is necessary. This reduces the complexity in running completions, electric control line protection, tubing hanger penetration, and the overall cost to the customer.

[0026] FIG. **1**A illustrates a well system **100** designed, manufactured and/or operated according to one or more embodiments of the disclosure. The well system **100**, in at least one embodiment, includes an offshore platform **110** connected to a first downhole device **170** (e.g., first SSSV, such as a TRSV) insert within a wellbore **130** (e.g., the wellbore extending through one or more subterranean formations) and a second downhole device **180** (e.g., second SSSV, such as a WLRSV) insert within the wellbore **130** via an electric control line **120** (e.g., primary electric control line, single electric control line, TEC, etc.). In at least one embodiment, the second downhole device **180** is an electrical connection for a WLRSV. For example, the electrical connection may be an inductive coupling, a capacitive coupling, or a conductive coupling with direct electrical contact, among others. An annulus **150** may be defined between walls of the wellbore **130** (e.g., extending through a subterranean formation) and a conduit **140**. A wellhead **160** may provide a means to hand off and seal conduit **140** against the wellbore **130** and provide a profile to latch a subsea blowout preventer to. Conduit **140** may be coupled to the wellhead **160**. Conduit **140** may be any conduit such as a casing, liner, production tubing, or other oilfield tubulars disposed in a wellbore. The first downhole device **170**, or at least a portion thereof, may be interconnected with the conduit **140** (e.g., disposed in line with the conduit **140**) and positioned in the wellbore **130**. The second downhole device **180**, or at least a portion thereof, may be interconnected with the conduit **140** (e.g., positioned within an ID or OD of the conduit **140**) and positioned in the wellbore **130**. In the illustrated embodiment, the second downhole device **180** is illustrated uphole of the first downhole device **170** (e.g., a portion of it being run-in-hole with the first downhole device **170** and another portion of it being run-in-hole after the first downhole device **170** has failed), but other embodiments may exist wherein the second downhole device **180** is located downhole of the first downhole device **170**.

[0027] The electric control line **120** may extend into the wellbore **130** and may be connected to the first downhole device **170** and the second downhole device **180**. The electric control line **120** may provide actuation power to the first downhole device **170** and the second downhole device **180**. As will be described in further detail below, power may be provided to first downhole device **170** or the second downhole device **180** to actuate or de-actuate the first downhole device **170** or the second downhole device **180**. Actuation may comprise opening the first downhole device **170** or the second downhole device **180** to provide a flow path for subsurface production fluids to enter conduit **140**, and de-actuation may comprise closing the first downhole device **170** or the second downhole device **180** to close a flow path for subsurface production fluids to enter conduit **140**. While the embodiment of FIG. **1**A illustrates only the first downhole device **170** and the second downhole device **180**, other embodiments exist wherein more than two downhole devices according to the disclosure are used.

[0028] In accordance with one embodiment of the disclosure, the well system **100** may further include a switch system **190***a* positioned between the electric control line **120** and each of the first downhole device **170** and the second downhole device **180**. The switch system **190***a*, as discussed above, is configured to switch the incoming power from the electric control line **120** between the first downhole device **170** and the second downhole device **180**, depending on which of the first downhole device **170** or the second downhole device **180** that the operator intends to operate (e.g., actuate). In at least one embodiment, the first downhole device **170** includes a first electrical devices (e.g., electromagnetic coils, electric motor or pump, piezoelectric actuator, solenoid valve, etc.) and the second downhole device **180** includes a second electrical devices (e.g., electromagnetic coils, electric motor or pump, piezoelectric actuator, solenoid valve, etc.), and the switch system **190***a* is configured to switch the incoming power from the electric control line **120** between the first electrical device of the first downhole device **170** and the second electrical device of the second downhole device **180**.

[0029] While the embodiment of FIG. **1**A employs a single electric control line **120** and the switch system **190***a*, other embodiments of the disclosure could use two or more different electric control lines with or without the switch system **190***a*. Although the well system **100** is depicted in FIG. **1**A as an offshore well system, one of ordinary skill should be able to adopt the teachings herein to any type of well, including onshore or offshore. In the embodiment of FIG. **1**A, the first downhole device **170** is a TRSV, and the second downhole device **180** is a WLRSV.

[0030] Turning to FIGS. **1**B and **1**C, illustrated is one embodiment of a switch system **190***b* designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system **100** of FIG. **1**A. The switch system **190***b*, in the illustrated embodiment, is a mechanical switch system. In the illustrated embodiment, the switch system **190***b* includes a mechanically activated switch **191**, the mechanically activated switch **191** having an input thereof coupled to the electric control line **120**, and a first output thereof coupled to the first downhole device **170** and a second output thereof coupled to the second downhole device **180**. Accordingly, the mechanically activated switch **191** switches the input power from the electric control line **120** between the first downhole device **170** (e.g., FIG. **1**B) and the second downhole device **180** (FIG. **1**C), as necessary.

[0031] While a number of different embodiments for mechanical switch systems may be used, in the illustrated embodiment, a sliding sleeve **172** of the first downhole device **170** includes a magnetic target **174** thereon. Similarly, the switch system **190***b* includes a related magnetic target **192** therein, for example coupled to the mechanically activated switch **191** (e.g., two or more magnetic features). In at least one embodiment, at least one of the magnetic target **174** or the magnetic target **192** is a magnet (e.g., permanent magnet or electromagnet). Furthermore, the switch system **190***b* may include an insulator **193** separating the first output and the second output. Accordingly, the related magnetic target **192** will couple with (e.g., decouple from) the magnetic target **174** to switch the power between the first downhole device **170** and the second downhole device **180**, in this instance as the sliding sleeve **172** moves, as shown in FIGS. **1**B and **1**C. In at least one embodiment, the sliding sleeve **172** is configured to move when the second downhole device **180** is being run-in-hole. Again, while one or more magnetic targets **174** are illustrated in FIGS. **1**B and **1**C for shifting the switch, in one or more other embodiments the switches are directly shifted as opposed to magnetically shifted.
[0032] While not illustrated in FIGS. **1**B and **1**C, another embodiment may exist wherein a reed switch is employed to switch between the first downhole device **170** and the second downhole device **180**. In such an embodiment, one or more of the magnetic targets **192** could be exchanged for a reed switch. Thus, as the magnetic target **174** passes over the reed switch, the reed switch will switch the power between the first downhole device **170** and the second downhole device **180**. In at least one embodiment, ones of the one or more reed switches are single pole single-throw reed switches and/or single pole double-throw reed switches. Those skilled in the art appreciate how such reed switches would be configured to achieve the desires stated herein.
[0033] Turning to FIGS. **1**D and **1**E, illustrated is one embodiment of a switch system **190***d* designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system **100** of FIG. **1**A. The switch system **190***d*, in the illustrated embodiment, is an electrical switch system, for example including an electrically activated switch. In the illustrated embodiment, the switch system **190***d* includes two or more oppositely oriented diodes **195***a*, **195***b* coupled between the electric control line **120** and each of the first downhole device **170** and the second downhole device **180**, respectively. The term "diode," as used herein, includes all electronics that have an asymmetric conductance, including semiconductor diodes, thermionic diodes, and multichip modules that have asymmetric conductance. Thus, for example, if a positive voltage is applied to the electric control line **120**, the first diode **195***a* would allow the current **197** to pass therethrough and thus would establish a closed circuit, and therefore the first downhole device **170** would be powered. However, the second diode **195***b* would not allow the current **197** to pass there through and thus would establish an open circuit, and thus the second downhole device **180** would not be powered. In contrast, if a negative voltage is applied to the electric control line **120**, the first diode **195***a* would not allow the current **197** to pass therethrough and thus would establish an open circuit, and therefore the first downhole device **170** would not be powered. However, the second diode **195***b* would allow the current **197** to pass therethrough and thus would establish a closed circuit, and thus the second downhole device **180** would be powered. Thus, by toggling the voltage between a positive voltage (e.g., preset positive voltage) and a negative voltage (e.g., preset negative voltage), the switch system **190***d* powers different ones of the first downhole device **170** and the second downhole device **180**.
[0034] Turning to FIGS. **1**F and **1**G, illustrated is one embodiment of a switch system **190***f* designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system **100** of FIG. **1**A. The switch system **190***f*, in the illustrated embodiment, is an electrical switch system. In the illustrated embodiment, the switch system **190***f* includes low frequency/high frequency filters coupled between the electric control line **120** and the first downhole device **170**, and/or high frequency/low frequency filters coupled between the electric control line **120** and the second downhole device **180**. In this scenario, the low frequency

filters would be configured to pass a low frequency signal of a power source (e.g., and block the high frequency signal of a power source), and the high frequency filters would be configured to pass the high frequency signal of the power source (e.g., and block the low frequency signal of the power source). Accordingly, by switching the frequency of the power source, one of the first downhole device **170** or second downhole device **180** will receive power, while the other of the second downhole device **180** or the first downhole device **170** will not receive power.

[0035] In the embodiment of FIGS. **1**F and **1**G, a low frequency filter **196***a* surrounds the first downhole device **170** and a high frequency filter **196***b* surrounds the second downhole device **180**. Accordingly, as shown in FIGS. **1**F and **1**G, the low frequency signal will only power the first downhole device **170** and the high frequency signal will only power the second downhole device **180**. Thus, by switching the frequency of the power source, different ones of the first downhole device **170** and second downhole device **180** will be powered.

[0036] It should be noted that the phrases “low frequency signal” and “high frequency signal” are relative to one another and not limited by any specific values. Nevertheless, in at least one embodiment, the low frequency signal is less than 100 Hz and the high frequency signal is greater than 100 Hz, and in even yet another embodiment greater than 10,000 Hz. In even yet another embodiment, the frequency of the high frequency signal is at least 50% higher (e.g., at least 50% more cycles per second) than the frequency of the low frequency signal. In even yet another embodiment, a DC signal is a subset of a low frequency signal.

[0037] It should additionally be noted that the phrase “frequency filter” includes all known or hereafter discovered frequency filters that could be used for the intended purpose disclosed herein. For example, the frequency filter could be a linear continuous-time filter, such as an elliptic filter, a Butterworth filter, or a Chebyshev filter, among others. The frequency filter can also be an analog filter or a digital filter, and can be a passive or active filter. In one example embodiment, the frequency filter is a passive analog filter. In even another embodiment, the frequency filter may include nonlinear electrical components, such as one or more electrical switch (e.g., like a field effect transistor or FET) and AC to DC power converters. In some embodiment, the frequency filter will induce the electrical switch to open or to close based on the frequency content of the input signal. In another embodiment, the output from the high-pass frequency filter is converted to a DC signal with an AC to DC converter. In other words, in at least this one embodiment the electrical power will only be delivered when the input signal is a high frequency signal. However, the electrical power that is delivered to the load **170**, in this embodiment, will consist of DC power.

[0038] In at least one embodiment, the first downhole device **170** is a sensor and the second downhole device **180** is a safety valve, such as a WLRSV. In at least this one embodiment, there is a desire to still power and/or communicate with the sensor even if the second downhole device **180** (e.g., WLRSV) is installed, and thus a frequency filter (e.g., high frequency filter) could be installed with the sensor. Accordingly, in this one embodiment, a first signal including DC power would be used to power the sensor, and when the second downhole device **180** (e.g., WLRSV) is installed, the first signal would be switched to a second signal including the DC power and AC power, such that the second downhole device **180** (e.g., WLRSV) and the sensor are powered. This approach might be used when separate power cables are employed for the TRSV and the sensor (e.g., downhole pressure/temperature sensor), and there is a need to drop in the WLRSV and provide power to it. Greater reliability may be achieved in this embodiment, given the fact that the second downhole device **180** (e.g., WLRSV) receives power along the same electric control line as the sensor.

[0039] It should further be noted that while the embodiment of FIGS. **1**F and **1**G employ only two downhole devices (e.g., first downhole device **170** and second downhole device **180**), and thus two frequency filters (e.g., low frequency filter **196***a* and high frequency filter **196***b*), other embodiments may exist wherein more than two downhole devices and more than two frequency filters are employed. For example, another embodiment might exist wherein the downhole tool includes first, second and third downhole devices, along with a low frequency filter, medium frequency filter and high frequency filter to achieve the same purpose as disclosed above. This idea could be expanded to any number of downhole devices and frequency filters.

[0040] Turning to FIGS. **1**H and **1**I, illustrated is one embodiment of a switch system **190***h* designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system **100** of FIG. **1**A. In the embodiment of FIGS. **1**H and **1**I, ones (e.g., pairs) of low frequency filters **196***a* surround the second downhole device **180** and ones (e.g., pairs) of high frequency filters **196***b* surround the first downhole device **170**. Accordingly, as shown in FIGS. **1**H and **1**I, a low frequency signal will only power the second downhole device **180** and a high frequency signal will only power the first downhole device **170**. Thus, by switching the frequency of the power source, ones of the first downhole device **170** and second downhole device **180** will be powered. Furthermore, while the embodiment of FIGS. **1**H and **1**I employ ones (e.g., pairs) of low frequency filters **196***a* surrounding the first downhole device **170** or high frequency filters **196***b*, and ones (e.g., pairs) of high frequency filters **196***b* or low frequency filters **196***a* surrounding the second downhole device **180**, other embodiments exist wherein a single low frequency filter **196***a* and/or a single high frequency filter **196***b* is positioned on one side or the other of the first downhole device **170** or second downhole device **180** (e.g., as shown in FIGS. **1**F and **1**G). Furthermore, not all embodiments require both the high frequency filter **196***b* and low frequency filter **196***a*, and thus certain circumstances may arise wherein a single frequency filter (e.g., either the high frequency filter **196***b* or the low frequency filter **196***a*) is employed, but not both. For example, the first output may be coupled to the first electrical component of the first downhole device via the frequency filter or the second output may be coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to switch power between the electric control line and the first downhole device or the electric control line and the second downhole device based upon switching a signal of the power source.

[0041] It should be noted that the embodiments of FIGS. **1**F through **1**I employ electromagnetic coupling to power the first downhole device **170** and the second downhole device **180**. Nevertheless, other embodiments could be employed wherein direct electrical coupling powers the first downhole device **170** and the second downhole device **180**. In yet another embodiment, a combination of electromagnetic coupling and direct coupling could be employed to power the first downhole device **170** and the second downhole device **180**.

[0042] Turning to FIGS. **1**J and **1**K, illustrated is one embodiment of a switch system **190***j* designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might be used in the well system **100** of FIG. **1**A. The switch system **190***j* contains a magnetically activated switch **198**. In one embodiment, the magnetically activated switch **198** is a reed switch, as shown in FIGS. **1**J and **1**K. When there is no magnetic field being subjected to the magnetically activated switch **198**, such as shown in FIG. **1**J, then the contact **199** in the reed switch is biased (e.g., inherently biased) towards an electrical connection with the first downhole device **170**, and thus power (e.g., electrical current) can flow to that tool. When there is a magnetic field being subjected to the magnetically activated switch **198**, such as shown in FIG. **1**K, then the contact **199** in the reed switch is biased (e.g., mechanically biased) towards an electrical connection with the second downhole device **180**, and thus power (e.g., electrical current) can flow to that tool. For example, in FIG. **1**K the magnetic target **174** creates a magnetic attraction that pulls the contact **199** towards an electrical connection with the second downhole device **180** and thus power (e.g., electrical current) can flow to that tool. The magnetically activated switch **198** can employ first and second reed switches rather than the double throw switch that is shown, wherein the second reed switch is configured to work in conjunction with the first reed switch to switch power between the electric control line and the first downhole device and the electric control line and the second downhole device. One of the advantages of the reed switch is that it is a mechanically activated switch and contains no electronics. As an alternative embodiment, the magnetically activated switch **198** could be a tunnel magneto-resistance (TMR) switch. A TMR switch contains a magnetic tunnel junction where the resistance of the junction varies with magnetic field. The TMR switch varies between high resistance (open switch) and low resistance (closed switch) with applied magnetic field.

[0043] Turning to FIG. **1**L, illustrated is a chart illustrating various different ways that an operator may provide power to the TRSV and/or WLRSV, including using a single electric control line, two dedicated electric control lines, a single electric control line with a switch, as well as a single electric control line with low/high frequency filters.

[0044] Turning to FIGS. **2**A through **2**F illustrated is one embodiment of a downhole device, including a safety valve **200** designed, manufactured and/or operated according to one or more embodiments of the disclosure, as might employ the first, second and third portions of the WLRSV, as discussed above. FIGS. **2**A through **2**C illustrate different views of the safety valve **200** in a first closed position, its unpowered electromagnetic assembly and magnetic target decoupled from one another. FIG. **2**D illustrates the safety valve **200** of FIGS. **2**A through **2**C in a second closed position with power (DC power in this embodiment) supplied to the electromagnetic assembly, thereby coupling the electromagnetic assembly and the magnetic target together. FIG. **2**E illustrates the safety valve **200** of FIG. **2**D now in an open position, the powered (DC powered) electromagnetic assembly and magnetic target remaining magnetically coupled (e.g., fixedly coupled) with one another. FIG. **2**F illustrates the safety valve **200** of FIG. **2**E after power (DC power) has been cut to the electromagnetic assembly, and thus the safety valve **200** returns to the first closed position. In yet another embodiment, the safety valve **200** may be indirectly moved back to the first closed position, for example if an electrical logic circuit determines that the electrical power has been interrupted and initiates a closing of the safety valve **200**.

[0045] Referring initially to FIGS. **2**A through **2**C, the safety valve **200** is illustrated in the first closed position. The safety valve **200**, in one or more embodiments, may include an outer housing **224** (e.g., tubular housing, wellbore tubing, etc.) containing a central bore **225** therein, wherein components of the safety valve **200** may be disposed within the central bore **225**. An upper valve assembly **234** (e.g., also the axially fixed ferromagnetic portion in this embodiment) may be attached to the outer housing **224**, and may further include one or more sealing elements **223**, such that fluid communication from a lower section **202** to an upper section **203** is prevented.

[0046] A sleeve **226** may be attached between the upper valve assembly **234** and a lower valve assembly **216**. A bore flow management actuator **240** may be disposed within the sleeve **226**. The bore flow management actuator **240** may include a translating sleeve **222** and a flow tube main body **208**. A flow path **214** may be defined by an interior of the flow tube main body **208**. As illustrated in FIGS. **2**A through **2**C, the flow path **214** may extend from an interior of a conduit **206** through an interior of the flow tube main body **208**. As will be discussed in further detail below, when the safety valve **200** is in an open position, the flow path **214** may extend from an interior of the conduit **206** through an interior of the flow tube main body **208** and further into the lower section **202**.

[0047] The safety valve **200** may further include a power spring **210** disposed between the lower valve assembly **216** and a translating sleeve shoulder **218**. As illustrated in FIGS. **2**A through **2**C, the translating sleeve shoulder **218** and a flow tube shoulder **232** may be in contact when the safety valve **200** is in the first closed position. The power spring **210** may provide a positive spring force against the translating sleeve shoulder **218**, which may keep the flow tube main body **208** in a first position. The power spring **210** may also provide a positive spring force to return the flow tube main body **208** and the translating sleeve **222** to the first position (e.g., from a second position), as will be explained below.

[0048] The safety valve **200** may further include a nose spring **212** disposed between a translating sleeve assembly **230** and the flow tube shoulder **232**. The translating sleeve assembly **230** may be disposed between and attached to a piston **220** and the translating sleeve **222**. The power spring **210** and the nose spring **212** are depicted as coil springs in FIGS. **2**A through **2**F. However, the power spring **210** and the nose spring **212** may comprise any kind of spring and remain within the scope of the present disclosure, such as, for example, coil springs, wave springs, or fluid springs, among others.

[0049] In the illustrated embodiment, the translating sleeve assembly **230** may allow a force applied to a distal end of the piston **220** to be transferred into the translating sleeve **222**. A force may be applied to the distal end of the piston **220** by way of fluid communication from a channel **228** through an orifice **242**. A force applied to the piston **220** may move the translating sleeve **222** from a first position to a second position. The nose spring **212** may provide a positive spring force against the translating sleeve assembly **230** and the flow tube shoulder **232**, which may return the translating sleeve **222** from the second position to the first position, as will be discussed in greater detail below.

[0050] In the first closed position, the translating sleeve **222** and the flow tube main body **208** are positioned such that the translating sleeve shoulder **218** and the flow tube shoulder **232** are in contact and the power spring **210** and the nose spring **212** are in an extended position. In the first closed position, the translating sleeve **222** may be referred to as being in a first position and the flow tube main body **208** may be referred to as being in a first position.

[0051] In at least one embodiment, the bore flow management actuator **240** is configured to slide from a first initial state to a first subsequent state to move a valve closure mechanism **204** between a first closed state and a first open state. In the first closed state, the valve closure mechanism **204** may isolate the lower section **202** from the flow tube main body **208**. When the valve closure mechanism **204** is in a first closed state, as in FIGS. **2**A through **2**C, the valve closure mechanism **204** may prevent formation fluids and pressure from flowing into the flow tube main body **208** from the lower section **202**. Although FIGS. **2**A through **2**C illustrate the valve closure mechanism **204** as a flapper valve, the valve closure mechanism **204** may be any suitable type of valve such as a flapper type valve, a linear stopper type valve, or a ball type valve, for example. As will be illustrated in further detail below, the valve closure mechanism **204** may be actuated into a first open state to allow formation fluids to flow from the lower section **202** through the flow path **214** (e.g., defined by the lower section **202**, an interior of the flow tube main body **208** and an interior of the conduit **206**).

[0052] When the safety valve **200** is in the first closed position, no amount of differential pressure across the valve closure mechanism **204** will allow formation fluids to flow from the lower section **202** into the flow path **214**. In the first closed position, the safety valve **200** will only allow fluid flow from conduit **206** into the lower section **202**, but not from the lower section **202** into the conduit **206**. In the instance that pressure in the conduit **206** is increased, the valve closure mechanism **204** will remain in the closed position until the pressure in the conduit **206** is increased above the pressure in the lower section **202** plus the closing pressure provided by the valve closure mechanism spring **205**, sometimes referred to herein as valve opening pressure. When the valve opening pressure is reached, the valve closure mechanism **204** may open and allow fluid communication from the conduit **206** into the lower section **202**. In this manner, treatment fluids such as surfactants, scale inhibitors, hydrate treatments, and other suitable treatment fluids may be introduced into the subterranean formation. The configuration of the safety valve **200** may allow treatment fluids to be pumped from a surface, such as a wellhead, into the subterranean formation without actuating a control line or balance line to open the valve. Once pressure in the conduit **206** is decreased below the valve opening pressure, the valve closure mechanism spring **205** will return the valve closure mechanism **204** to the closed position, and thus flow from the conduit **206** into the lower section **202** will cease. When the valve closure mechanism **204** has returned to the closed position, flow from the lower section **202** into the flow path **214** will be prevented. Should a pressure differential across the valve closure mechanism **204** be reversed, such that pressure in the lower section **202** is greater than a pressure in the conduit **206**, the valve closure mechanism **204** will remain in a closed position, such that fluids in the lower section **202** are prevented from flowing into the conduit **206**.

[0053] In the illustrated embodiment, the safety valve **200** includes a first portion **250**, a second portion **260** (e.g., the second portion **260** may include those features disclosed in the paragraph above, for example those feature located between the upper valve assembly **234** and the valve closure mechanism **204**, and specifically the bore flow management actuator **240** and the valve closure mechanism **204**), and a third portion **270**. As indicated above, in at least one embodiment, the first portion **250** has a first portion minimum inside diameter (ID.sub.1) and is run-in-hole with the TRSV, and the second portion **260** and the third portion **270** are run-in-hole after the TRSV is no longer working properly and/or has failed. For example, in at least one embodiment, the second portion **260** has a second portion maximum outside diameter (OD.sub.2), the second portion maximum outside diameter (OD.sub.2) being less than the first portion minimum inside diameter (ID.sub.1) such that the second portion **260** may be run-in-hole after the first portion **250**. Furthermore, the third portion **270** may be run-in-hole in a separate step after the second portion **260**.

[0054] In one or more embodiments, the first portion **250** includes a pocket **251** formed therein. In at least one embodiment, the pocket **251** has an electromagnetic assembly **254** (e.g., including one or more coils) located therein. The one or more coils, in one or more embodiments, may include an insulated electrical wire that makes loops around a common axis in order to produce a magnetic field when a current passes through the wire. The number of loops may vary, but in at least one embodiment the number of loops is between 10 and 500,000, if not between 100 and 100,00. In one or more other embodiments, the pocket **251** has one or more radial outer magnetic targets **257** and a magnetic target **258**, for example positioned on a sliding sleeve **256**, located therein. For example, in at least one embodiment the one or more radial outer magnetic targets **257** and the magnetic target **258** are configured to slide within the pocket **251**, for example being placed on or in a sliding sleeve (e.g., not shown). Accordingly, the one or more radial outer magnetic targets **257** and the magnetic target **258** may slide within the pocket relative to the electromagnetic assembly **254** (e.g., fixed electromagnetic assembly). In one or more embodiments, a fluid isolation sleeve **252** isolates the electromagnetic assembly **254**, one or more radial outer magnetic targets **257**, and the magnetic target **258** from fluid and/or debris within the wellbore. The fluid isolation sleeve **252** may be ported to allow pressure balancing. In one embodiment, the fluid isolation sleeve **252** is mechanically connected to the electromagnetic assembly **254**. In one embodiment, the fluid isolation sleeve **252** is mechanically connected to the electromagnetic assembly **254**. In at least one embodiment, the fluid isolation sleeve **252** is a fixed fluid isolation sleeve, and thus does not readily move once positioned downhole. For example, the fluid isolation sleeve **252** could comprise a composite, a plastic, a ceramic, aluminum, stainless steel, or another non-ferromagnetic material. In yet another embodiment, the fluid isolation sleeve **252** could comprise a ferromagnetic material, but would need to be sufficient thin as to not draw too much of the magnetic force generated by the electromagnetic assembly **254** from achieving its intended use, as discussed below.

[0055] In one or more embodiments, the second portion **260** includes the flow tube **208** and the valve **204**, and may be located below the first portion **250** (e.g., below the fluid isolation sleeve **252**, and the electromagnetic assembly **254**). The second portion **260** may additionally include the power spring **210** and/or nose spring **212**, as will be further discussed below.

[0056] In one or more other embodiments, the third portion **270** includes a mechanical connecting apparatus **272**, the mechanical connecting apparatus including one or more radial inner magnetic targets **274** associated therewith. The one or more radial inner magnetic targets **274**, in this embodiment, are configured to magnetically couple with the one or more radial outer magnetic targets **257** of the first portion **250** (e.g., through the fluid isolation sleeve **252**). The mechanical connecting apparatus **272**, in at least one embodiment, axially couples the one or more radial inner magnetic targets **274** of the third portion and the flow tube **208** of the second portion **260**, and thus the one or more radial outer magnetic targets **257** of the first portion **250** with the flow tube **208** of the second portion **260**.

[0057] With reference to FIG. **2**D the safety valve **200** is illustrated in a second closed position. In the second closed position, the translating sleeve **222** may be displaced from the first position to a second position, which is relatively closer in proximity to the valve closure mechanism **204**. The flow tube main body **208** may remain in the first position, or alternatively only slightly downhole from the first position. When the safety valve **200** is in the second closed position, both the power spring **210** and the nose spring **212** may be in a compressed state.

[0058] To move the translating sleeve **222** to the second position, differential pressure across the valve closure mechanism **204** may be increased by lowering the pressure in the conduit **206** or increasing pressure in the lower section **202**. Lowering pressure in the conduit **206** or increasing pressure in the lower section **202** may cause fluid from the lower section **202** to flow through the channel **228** defined between the sleeve **226** and the outer housing **224** into the orifice **242**. The orifice **242** may allow fluid communication into the piston tube **244**, whereby fluid pressure may act on the proximal end of the piston **220**. The force exerted by fluid pressure on the proximal end of the piston **220** may displace the piston **220** towards the valve closure mechanism **204** by transferring the force through the piston **220**, the translating sleeve assembly **230**, and the translating sleeve shoulder **218**. The nose spring **212** may provide a spring force against the flow tube shoulder **232** and the translating sleeve assembly **230**, and the power spring **210** may provide a spring force against the translating sleeve shoulder **218** and the lower valve assembly **216**.

[0059] Although not illustrated in FIGS. **2**A through **2**F, the flow tube main body **208** may include channels that allow pressure and/or fluid communication between the flow path **214** and an interior of the sleeve **226**. Collectively the spring forces from the power spring **210** and the nose spring **212** may resist the movement of the piston **220** until the differential pressure across the valve closure mechanism **204** is increased beyond the spring force provided from the power spring **210** and the nose spring **212**. Increasing differential pressure may include decreasing pressure in the conduit **206** such that the pressure in the lower section **202** is relatively higher than the pressure in the conduit **206**. When the differential pressure across the valve closure mechanism **204** is increased, the differential pressure across the piston **220** also increases. When the differential pressure across the valve closure mechanism **204** is increased beyond the spring force provided by the nose spring **212** and the power spring **210**, the nose spring **212** and the power spring **210** may compress and allow the translating sleeve **222** to move into the second position. Differential pressure across the valve closure mechanism **204** may be increased by pumping fluid out of the conduit **206**, for example. In the instance that the lower section **202** is fluidically coupled to a non-perforated section of pipe or where there is a plug in a conduit **206** fluidically coupled to the lower section **202** that prevents pressure being transmitted from the lower section **202** to the piston **220**, a pressure differential across the valve closure mechanism **204** may be induced through pipe swell.

[0060] In the second closed position, the safety valve **200** remains safe as no fluids from the lower section **202** can flow into the flow path **214**. In the second closed position no amount of differential pressure across the valve closure mechanism **204**, the differential pressure being relatively higher pressure in the lower section **202** and relatively lower pressure in the conduit **206**, should cause the valve closure mechanism **204** to open to allow fluids from the lower section **202** to flow into the flow path **214**, as the pressure from the lower section **202** is acting on the valve closure mechanism **204**. If pressure is increased in the conduit **206**, the differential pressure across the valve closure mechanism **204** decreases and the translating sleeve **222** may move back to the first position illustrated in FIGS. **2**A through **2**C. Unlike conventional safety valves that generally require a control line to supply pressure to actuate a piston to move a translating sleeve, the safety valve **200** may only require pressure supplied by the wellbore fluids in the lower section **202** to move the

translating sleeve.
[0061] With continued reference to FIG. **2**D, the piston **220** may be fixedly attached to the translating sleeve assembly **230**. Although illustrated as a single piston in FIGS. **2**A through **2**F, the piston **220** may comprise a plurality of pistons and remain within the scope of the disclosure.
[0062] As illustrated, when the translating sleeve **222** is moved from the first position to the second position, the piston **220** may also be moved. Before, during or after the translating sleeve **222** is allowed to come to the second position as described above, the electromagnetic assembly **254** may be powered on. The powering on of the electromagnetic assembly **254** may cause the electromagnetic assembly **254** to hold the magnetic target **258**, which is axially fixed to the one or more radial outer magnetic targets **257** (e.g., via the sliding sleeve **256**), and the one or more radial inner magnetic targets **274** of the third portion **270** in their axial downhole position. As the one or more radial inner magnetic targets **274** are rigidly coupled to the flow tube **208** (e.g., via the mechanical connecting apparatus **272**), the flow tube **208** is also held in its axial downhole position, as shown in FIG. **2**D.
[0063] In FIGS. **2**A through **2**F, the electromagnetic assembly **254** is depicted as one coil circumscribing the tubular, but there may be any number of coils in any orientation to fix the sliding sleeve **226**, and thus bore flow management actuator **240** in place. The electromagnetic assembly **254** may apply a force in a substantially radial or axial direction, for example. The force applied by the electromagnetic assembly **254** may be any amount of force, including but not limited to, a force in a range of about 45 Newtons to about 45000 Newtons. The electromagnetic assembly **254** may provide a means to hold the sliding sleeve **226** and the bore flow management actuator **240** at any well depth. Hydraulic systems used in previous wellbore safety valves generally require control and balance lines to actuate and hold a valve open which may have pressure limitations. The limitations experienced by hydraulic systems may be overcome by using the electromagnetic assembly **254** described herein, as only well pressure is required to open the safety valve **200**. Again, when the translating sleeve **222** is in the second position either when the electromagnetic assembly **254** is switched on or switched off, no amount of differential pressure across the valve closure mechanism **204** will open the valve closure mechanism **204**, the differential pressure being a pressure difference between a relatively higher pressure in the section **202** and a relatively lower pressure in the conduit **206**.
[0064] With reference to FIG. **2**E, the safety valve **200** is illustrated in an open position. When the safety valve **200** is in the open position, the translating sleeve **222** may be fixed in place in the second position, as in FIGS. **2**D and **2**E, through the force provided by the electromagnetic assembly **254**, the force being transferred through the mechanical connecting apparatus **272** to the bore flow management actuator **240**, for example via the translating sleeve **222**. The flow tube main body **208** is illustrated as being axially shifted from the first position illustrated in FIGS. **2**A through **2**D to a second position in FIG. **2**E. When the flow tube main body **208** is in the second position, the flow tube shoulder **232** and the translating sleeve shoulder **218** may be in contact and the flow tube main body **208** may have displaced the valve closure mechanism **204** into an open position. The nose spring **212** may be in an uncompressed state, while the power spring **210** may be in a compressed state.
[0065] The flow tube main body **208** may be moved from the first position to the second position when the translating sleeve **222** is fixed in place in the second position by the electromagnetic assembly **254**, as described above. When the translating sleeve **222** is fixed in the second position through the force provided by the electromagnetic assembly **254**, the nose spring **212** may provide a positive spring force against the flow tube shoulder **232** and the translating sleeve assembly **230**. The positive spring force from the nose spring **212** may be transferred through the flow tube main body **208** into the valve closure mechanism **204**. The flow tube main body **208** will not move to the second position until differential pressure across the valve closure mechanism **204** exists and the translating sleeve **222** is fixed in position. The differential pressure may be decreased by pumping

into the conduit **206**, thereby increasing the pressure in the conduit **206**. The pressure may be increased in the conduit **206** until the differential pressure across the valve closure mechanism **204** is decreased to a point where the positive spring force from the nose spring **212** is greater than the differential pressure across the valve closure mechanism **204**. Thereafter, the nose spring **212** may extend and move the flow tube main body **208** into the second position by acting on the translating sleeve assembly **230** and the flow tube shoulder **232**, which are held in place via the electromagnetic assembly **254** and one or more other features. When the flow tube main body **208** is in the second position, fluids such as oil and gas in the lower section **202** may be able to flow into the flow path **214** and to a surface of the wellbore such as to a wellhead. Safety valve **200** may remain in the open position defined by the translating sleeve **222** being in the second position and the flow tube main body **208** being in the second position, as long as the electromagnetic assembly **254** remains powered on.

[0066] The safety valve **200** may be moved back to the first closed position, as illustrated in FIG. **2**F, by powering off the electromagnetic assembly **254**. As previously discussed, the electromagnetic assembly **254** may fix the one or more radial inner magnetic targets **274** and the flow tube **208** in place in the second position when the electromagnetic assembly **254** remains powered on. When the electromagnetic assembly **254** is powered off, the one or more radial inner magnetic targets **274** and the flow tube **208** may no longer be fixed in place. The electromagnetic assembly **254** may alternate the direction of the wrapping of the coil. Alternating the direction of the wrapping will result in alternating directions of the magnetic field. The result is multiple magnetic reversals in the coils of the electromagnetic assembly **254** that would magnetically connect with the one or more radial inner magnetic targets **274**. The power spring **210** may provide a positive spring force against the lower valve assembly **216**, translating the sleeve shoulder **218** and the flow tube shoulder **232** uphole. The positive spring force from the power spring **210** may axially displace the translating sleeve **222** to the first position and the flow tube main body **208** to the first position, thereby returning the safety valve **200** to the first closed position illustrated in FIGS. **2**A through **2**C, and **2**F. The positive spring force from the power spring **210** may also axially displace the one or more radial inner magnetic targets **274** to the position illustrated in FIGS. **2**A through **2**C, and **2**F, by transmitting the positive spring force through the mechanical connecting apparatus **272**.

[0067] Turning now to FIGS. **3**A through **3**D, illustrated are different views of a safety valve **300** designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure. The safety valve **300** of FIGS. **3**A through **3**D, is similar in many respects to the safety valve **200** of FIGS. **2**A through **2**C. Accordingly, like reference numbers have been used to illustrate similar, if not identical, features. FIGS. **3**A through **3**D illustrate the safety valve **300** in its operational state, thus each of the first portion **250**, the second portion **260** and the third portion **270** are coupled together and fixed within the tubular. For example, as illustrated, a latching mechanism **362** of the second portion **260** axially fixes the second portion **260** within the tubular. Furthermore, the mechanical connecting apparatus **272** of the third portion **270** axially fixes the one or more radial inner magnetic targets **274** of the third portion **270** to the bore flow management actuator **240** (e.g., translating sleeve assembly **230**) of the second portion **260**. Thus, any axial movement of the flow tube **208**, as discussed above, with result in a similar axial movement of the one or more radial inner magnetic targets **274**. Additionally, the electromagnetic assembly **254**, one or more radial outer magnetic targets **257**, magnetic target **258**, and sliding sleeve **256**, in the illustrated embodiment, are located in the pocket **251** in a safety valve sub **320**. Further to this embodiment, the fluid isolation sleeve **252** separates the electromagnetic assembly **254**, one or more radial outer magnetic targets **257**, magnetic target **258**, and sliding sleeve **256** located in the pocket **251**, from fluid within the wellbore, and thus from the mechanical connecting apparatus **272**.

[0068] Turning now to FIGS. **4**A through **9**D, illustrated are various different installation and/or operational states, each with various different views, of a safety valve **400** designed, manufactured and/or operated according to one or more alternative embodiments of the disclosure. The safety valve **400** of FIGS. **4**A through **9**D, is similar in many respects to the safety valve **300** of FIGS. **3**A through **3**D. Accordingly, like reference numbers have been used to illustrate similar, if not identical, features.

[0069] FIGS. **4**A through **4**D illustrate the safety valve **400** in its original run-in-hole state, and thus at this stage the safety valve **400** only includes a safety valve sub **420** coupled to the TRSV **410**. The safety valve sub **420**, in this embodiment, includes a first portion **250** (e.g., having the electromagnetic assembly **254**, sliding sleeve **256**, electromagnetic assembly **254**, one or more radial outer magnetic targets **257**, and magnetic target **258**, all of which are located in the pocket **251** and protected by the fluid isolation sleeve **252**). In contrast, FIGS. **5**A through **5**D illustrate the safety valve **400** after the TRSV **410** is no longer working properly and/or has failed. Accordingly, the safety valve **400** of FIGS. **5**A through **5**D additionally includes the second portion **260**, for example including the flow tube main body **208** and valve closure mechanism **204**.

[0070] Turning to FIGS. **6**A through **6**D, illustrated is the safety valve **400** of FIGS. **5**A through **5**D after inserting the third portion **270** therein. In the illustrated embodiment, the third portion **270** includes the mechanical connecting apparatus **272** and one or more radial inner magnetic targets **274**. The third portion **270**, in this embodiment, axially couples with the flow tube **208** of the second portion **260**. Accordingly, in the embodiment of FIGS. **6**A through **6**D, the third portion **270** is coupled to the flow tube **208** of the second portion **260**, and thus the one or more radial inner magnetic targets **274** are axially fixed to the flow tube **208**. Furthermore, the one or more radial inner magnetic targets **274** are magnetically coupled with the one or more radial outer magnetic targets **257**, for example across the fluid isolation sleeve **252**. The progression of FIGS. **4**A through **6**D illustrates how the safety valve **400** could be installed in accordance with one or more embodiments of the disclosure.

[0071] Turning now to FIGS. **7**A through **9**D, illustrated are certain embodiments how the safety valve **400** of FIGS. **6**A through **6**D could be operated. Specifically, FIGS. **7**A through **7**D illustrate the safety valve **400** of FIGS. **6**A through **6**D when the safety valve **400** has tubing pressure below the valve closure mechanism **204**. In this instance, the pressure below the valve closure mechanism **204** has compressed the power spring **210** and nose spring **212**, and in doing so may have slightly moved the flow tube **208** and associated mechanical connecting apparatus **272** downhole. For example, the flow tube **208** has moved downhole into contact with the valve closure mechanism **204**.

[0072] Turning to FIGS. **8**A through **8**D, illustrated is the safety valve **400** of FIGS. **7**A through **7**D after the pressure is balanced across the valve closure mechanism **204**, and thus the flow tube **208** is allowed to move axially down to open the valve closure mechanism **204**. At this stage, the electromagnetic assembly **254** and the magnetic target **258** are located proximate one another. If and/or when the electromagnetic assembly **254** is powered on, the electromagnetic assembly **254** will engage with the magnetic target **258**, and thus with the one or more radial outer magnetic targets **257** and associated one or more inner radial permanent magnets **274** axially hold the flow tube **208** in this open state.

[0073] Turning now to FIGS. **9**A through **9**D, illustrated is the safety valve **400** of FIGS. **8**A through **8**D after the electromagnetic assembly **254** loses power, and thus the power spring **210** of the second portion **260** pushes the flow tube main body **208** uphole, allowing the valve closure mechanism **204** to close.

[0074] Aspects disclosed herein include: [0075] A. A safety valve, the safety valve including: 1) a first portion, the first portion including a safety valve sub having an electromagnetic assembly, the first portion configured to be run-in-hole as part of wellbore tubing; 2) second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; and 3) third portion, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the electromagnetic assembly when the electromagnetic assembly is in an energized state to axially fix the bore flow management actuator in the first subsequent state [0076] B. A well system, the well system including: 1) a wellbore extending through one or more subterranean formations; 2) production tubing disposed in the wellbore; and 3) a safety valve disposed in the wellbore, the safety valve including: a) a first portion, the first portion including a safety valve sub having an electromagnetic assembly, the first portion configured to be run-in-hole as part of wellbore tubing; b) a second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; and c) a third portion, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the electromagnetic assembly when the electromagnetic assembly is in an energized state to axially fix the bore flow management actuator in the first subsequent state. [0077] C. A method, the method including: 1) positioning a first portion within a wellbore extending through one or more subterranean formations, the first portion including a safety valve sub having an electromagnetic assembly; 2) positioning a second portion within the wellbore, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state, the second portion configured to be run-in-hole after the first portion; and 3) positioning a third portion within the wellbore, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to be run-in-hole after the second portion to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the electromagnetic assembly when the electromagnetic assembly is in an energized state to axially fix the bore flow management actuator in the first subsequent state. [0078] D. A safety valve, the safety valve including: 1) a first portion, the first portion including a safety valve sub having an electromagnetic assembly; 2) a second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; 3) an axially fixed magnetic target portion, the axially fixed magnetic target portion and the electromagnetic assembly configured to create a magnetic flux when the electromagnetic assembly is energized; and 4) a third portion, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the axially fixed magnetic target portion via the magnetic flux to axially fix the bore flow management actuator in the first subsequent state. [0079] E. A well system, the well system including: 1) a wellbore extending through one or more subterranean formations; 2) production tubing disposed in the wellbore; and 3) a safety valve disposed in the wellbore, the safety valve including: a) a first portion, the first portion including a safety valve sub having an electromagnetic assembly; b) a second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; c) an axially fixed magnetic target portion, the axially fixed magnetic target portion and the electromagnetic assembly configured to create a magnetic flux when the electromagnetic assembly is energized; and d) a third portion, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the axially fixed magnetic target portion via the magnetic flux to axially fix the bore flow management actuator in the first subsequent state. [0080] F. A method, the method including: 1) positioning a first portion within a wellbore extending through one or more subterranean formations, the first portion including a safety valve sub having an electromagnetic assembly; 2) positioning a second portion within the wellbore, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; 3) positioning an axially fixed magnetic target portion within the wellbore, the axially fixed magnetic target portion and the electromagnetic assembly configured to create a magnetic flux when the electromagnetic assembly is energized; and 4) positioning a third portion within the wellbore, the third portion including a mechanical connecting apparatus having one or more magnetic targets associated therewith, the third portion configured to engage with at least a portion of the bore flow management actuator, the one or more magnetic targets configured to magnetically engage with the axially fixed magnetic target portion via the magnetic flux to axially fix the bore flow management actuator in the first subsequent state. [0081] G. A safety valve, the safety valve including: 1) a first portion, the first portion including a safety valve sub having a pocket therein, the pocket including an electromagnetic assembly, one or more radial outer magnetic targets, and a magnetic target located therein; 2) a second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; and 3) a third portion, the third portion including a mechanical connecting apparatus having one or more radial inner magnetic targets associated therewith and configured to magnetically engage with the one or more radial outer magnetic targets, wherein at least one of the one or more radial inner magnetic targets or the one or more radial inner magnetic targets are one or more permanent magnets, the third portion configured to engage with at least a portion of the bore flow management actuator such that when the bore flow management actuator moves to the first subsequent state the magnetic target moves proximate the electromagnetic assembly, and further wherein when the magnetic target is located proximate an energized electromagnetic assembly the bore flow management actuator is fixed in the first subsequent state. [0082] H. A well system, the well system including: 1) a wellbore extending through one or more subterranean formations; 2) production tubing disposed in the wellbore; and 3) a safety valve disposed in the wellbore, the safety valve including: a) a first portion, the first portion including a safety valve sub having a pocket therein, the pocket including an electromagnetic assembly, one or more radial outer magnetic targets, and a magnetic target located therein; b) a second portion, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; and c) a third portion, the third portion including a mechanical connecting apparatus having one or more radial inner magnetic targets associated therewith and configured to magnetically engage with the one or more radial outer magnetic targets, wherein at least one of the one or more radial inner magnetic targets or the one or more radial inner magnetic targets are one or more permanent magnets, the third portion configured to engage with at least a portion of the bore flow management actuator such that when the bore flow management actuator moves to the first subsequent state the magnetic target moves proximate the electromagnetic assembly, and further wherein when the magnetic target is located proximate an energized electromagnetic assembly the bore flow management actuator is fixed in the first subsequent state. [0083] I. A method, the method including: 1) positioning a first portion within a wellbore extending through one or more subterranean formations, the first portion including a safety valve sub having a pocket therein, the pocket including an electromagnetic assembly, one or more radial outer magnetic targets, and a magnetic target located therein; 2) positioning a second portion within the wellbore, the second portion including a valve closure mechanism and a bore flow management actuator, the bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the valve closure mechanism between a first closed state and a first open state; and 3) positioning a third portion within the wellbore, the third portion including a mechanical connecting apparatus having one or more radial inner magnetic targets associated therewith and configured to magnetically engage with the one or more radial outer magnetic targets, wherein at least one of the one or more radial inner magnetic targets or the one or more radial inner magnetic targets are one or more permanent magnets, the third portion configured to engage with at least a portion of the bore flow management actuator such that when the bore flow management actuator moves to the first subsequent state the magnetic target moves proximate the electromagnetic assembly, and further wherein when the magnetic target is located proximate an energized electromagnetic assembly the bore flow management actuator is fixed in the first subsequent state. [0084] J. A downhole tool, the downhole tool including: 1) a first downhole device; and 2) a switch system electrically coupled with the first downhole device, the switch system including: a) an input coupleable to a power source via an electric control line; b) an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and c) a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source. [0085] K. A well system, the well system including: 1) a wellbore extending through one or more subterranean formations; 2) production tubing disposed in the wellbore; 3) a downhole tool disposed in the wellbore, the downhole tool including: a) a first downhole device; and b) a switch system electrically coupled with the first downhole device, the switch system including: i) an input coupleable to a power source via an electric control line; ii) an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and iii) a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source. [0086] L. A method, the method including: 1) positioning a first downhole device in a wellbore; 2) positioning a second downhole device in the wellbore, wherein a switch system is coupled with the first and second downhole devices, the switch system including: a) an input coupleable to a power source via an electric control line; b) an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and c) a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source; and 3) switching a signal of the power source to operate ones of the first downhole device or the second downhole device.

[0087] Aspects A, B, C, D, E, F, G, H, I, J, K and L may have one or more of the following additional elements in combination: Element 1: wherein the electromagnetic assembly is located in a pocket in the safety valve sub. Element 2: wherein the first portion further includes a fluid isolation sleeve separating the electromagnetic assembly located in the pocket from the mechanical connecting apparatus. Element 3: wherein the fluid isolation sleeve is non-ferromagnetic. Element 4: wherein the fluid isolation sleeve is an axially fixed fluid isolation sleeve. Element 5: wherein the one or more magnetic targets are one or more permanent magnets. Element 6: wherein the bore flow management actuator includes a bore flow management actuator profile and the mechanical connecting apparatus includes a downhole mechanical connecting apparatus profile, and further wherein the downhole mechanical connecting apparatus profile is configured to couple with the bore flow management actuator profile to axially couple the at least a portion of the bore flow management actuator and the mechanical connecting apparatus. Element 7: wherein the bore flow management actuator includes a flow tube main body and a translating sleeve assembly including a translating sleeve, and further wherein the downhole mechanical connecting apparatus profile is configured to couple with the translating sleeve of the second portion to axially fix together the one or more magnetic targets and the translating sleeve. Element 8: wherein the second portion further includes a lower valve assembly located proximate the valve closure mechanism and an upper valve assembly positioned distal the valve closure mechanism, and further wherein the bore flow management actuator includes a power spring disposed between the lower valve assembly and a translating sleeve shoulder of the translating sleeve. Element 9: wherein the first portion is configured to be run-in-hole as part of wellbore tubing, the second portion is configured to be run-in-hole after the first portion, and the third portion is configured to be run-in-hole after the second portion. Element 10: wherein the electromagnetic assembly is located in a pocket in the safety valve sub. Element 11: wherein the first portion further includes a fluid isolation sleeve separating the electromagnetic assembly located in the pocket from the mechanical connecting apparatus. Element 12: wherein the fluid isolation sleeve is non-ferromagnetic. Element 13: wherein the fluid isolation sleeve is an axially fixed fluid isolation sleeve. Element 14: wherein the axially fixed magnetic target portion forms at least a portion of the second portion. Element 15: wherein the bore flow management actuator includes a bore flow management actuator profile and the mechanical connecting apparatus includes a downhole mechanical connecting apparatus profile, and further wherein the downhole mechanical connecting apparatus profile is configured to couple with the bore flow management actuator profile to axially couple the at least a portion of the bore flow management actuator and the mechanical connecting apparatus. Element 16: wherein the bore flow management actuator includes a flow tube main body and a translating sleeve assembly including a translating sleeve, and further wherein the downhole mechanical connecting apparatus profile is configured to couple with the translating sleeve of the second portion to axially fix together the one or more magnetic targets and the translating sleeve. Element 17: wherein the second portion further includes a lower valve assembly located proximate the valve closure mechanism and an upper valve assembly positioned distal the valve closure mechanism, and further wherein the bore flow management actuator includes a power spring disposed between the lower valve assembly and a translating sleeve shoulder of the translating sleeve. Element 18: wherein the first portion is configured to be run-in-hole as part of wellbore tubing, the second portion is configured to be run-in-hole after the first portion, and the third portion is configured to be run-in-hole after the second portion. Element 19: wherein the first portion is configured to be run-in-hole as part of wellbore tubing, the second portion is configured to be run-in-hole after the first portion, and the third portion is configured to be run-in-hole after the second portion. Element 20: wherein the one or more radial outer magnetic targets and the magnetic target are axially fixed relative to one another. Element 21: wherein at least one of the one or more radial outer magnetic targets and the magnetic target are located in a sliding sleeve in the pocket. Element 22: wherein the one or more radial outer magnetic targets are located in the sliding sleeve in the pocket. Element 23: wherein the safety valve sub has an uphole end and a downhole end, the one or more radial outer magnetic targets located more near the uphole end and the electromagnetic assembly located more near the downhole end, and further wherein the magnetic target is located between the one or more radial outer magnetic targets and the electromagnetic assembly. Element 24: wherein the first portion further includes a fluid isolation sleeve separating the electromagnetic assembly, one or more radial outer magnetic targets, and the magnetic target located in the pocket from the mechanical connecting apparatus. Element 25: wherein the fluid isolation sleeve is non-ferromagnetic. Element 26: wherein the fluid isolation sleeve is an axially fixed fluid isolation sleeve. Element 27: wherein the bore flow management actuator includes a bore flow management actuator profile and the mechanical connecting apparatus includes a downhole mechanical connecting apparatus profile, and further wherein the downhole mechanical connecting apparatus profile is configured to couple with the bore flow management actuator profile to axially couple the at least a portion of the bore flow management actuator and the mechanical connecting apparatus. Element 28: wherein the frequency filter is a first frequency filter, and further wherein the first output is coupled to the first electrical component of the first downhole device via the first frequency filter and the second output is coupleable to the second electrical component of the second downhole device via a second frequency filter, the first and second frequency filters configured to switch power between the electric control line and the first downhole device and the electric control line and the second downhole device based upon switching a frequency of the power source. Element 29: wherein the first frequency filter is a low frequency filter configured to pass a low frequency signal of the power source and block a high frequency signal of the power source, and the second frequency filter is a high frequency filter configured to pass the high frequency signal of the power source and block the low frequency signal of the power source. Element 30: wherein the first frequency filter is a high frequency filter configured to pass a high frequency signal of the power source and block a low frequency signal of the power source, and the second frequency filter is a low frequency filter configured to pass the low frequency signal of the power source and block the high frequency signal of the power source. Element 31: wherein the first downhole device further includes a first valve closure mechanism coupled to the first outer housing within the first central bore, and a first bore flow management actuator disposed in the first central bore, the first bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the first valve closure mechanism between a first closed state and a first open state. Element 32: wherein the first downhole device is a tubing retrievable safety valve (TRSV) and the second downhole device is a wireline retrievable safety valve (WLRSV). Element 33: wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) before or after the wireline retrievable safety valve (WLRSV) is insert within a wellbore. Element 34: wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) as the wireline retrievable safety valve (WLRSV) is being insert within a wellbore. Element 35: wherein the first frequency filter is one of a pair of first frequency filters surrounding the first downhole device, and the second frequency filter is one of a pair of second frequency filters surrounding the second downhole device. Element 36: wherein the first electrical component of the first downhole device is a first electromagnetic assembly and the second electrical component of the second downhole device is a second electromagnetic assembly. Element 37: wherein the first electrical component is an electric motor or pump, a piezoelectric actuator, or a solenoid valve. Element 38: wherein the output is a first output coupled to the first electrical component of the first downhole device and a second output coupled to the second electrical component of the second downhole device, and further wherein the first output is coupled to the first electrical component of the first downhole device via the frequency filter or the second output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to switch power between the electric control line and the first downhole device or the electric control line and the second downhole device based upon switching a signal of the power source.

[0088] Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

Claims

1. A downhole tool, comprising: a first downhole device; and a switch system electrically coupled with the first downhole device, the switch system including: an input coupleable to a power source via an electric control line; an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source.

2. The downhole tool as recited in claim 1, wherein the output is a first output coupled to the first electrical component of the first downhole device and a second output coupled to the second electrical component of the second downhole device, and further wherein the first output is coupled to the first electrical component of the first downhole device via the frequency filter or the second output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to switch power between the electric control line and the first downhole device or the electric control line and the second downhole device based upon switching a signal of the power source.

3. The downhole tool as recited in claim 2, wherein the frequency filter is a first frequency filter, and further wherein the first output is coupled to the first electrical component of the first downhole device via the first frequency filter and the second output is coupleable to the second electrical component of the second downhole device via a second frequency filter, the first and second frequency filters configured to switch power between the electric control line and the first downhole device and the electric control line and the second downhole device based upon switching a frequency of the power source.

4. The downhole tool as recited in claim 3, wherein the first frequency filter is a low frequency filter configured to pass a low frequency signal of the power source and block a high frequency signal of the power source, and the second frequency filter is a high frequency filter configured to pass the high frequency signal of the power source and block the low frequency signal of the power source.

5. The downhole tool as recited in claim 3, wherein the first frequency filter is a high frequency filter configured to pass a high frequency signal of the power source and block a low frequency signal of the power source, and the second frequency filter is a low frequency filter configured to pass the low frequency signal of the power source and block the high frequency signal of the power source.

6. The downhole tool as recited in claim 3, wherein the first downhole device further includes a first valve closure mechanism coupled to the first outer housing within the first central bore, and a first bore flow management actuator disposed in the first central bore, the first bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the first valve closure mechanism between a first closed state and a first open state.

7. The downhole tool as recited in claim 6, wherein the first downhole device is a tubing retrievable safety valve (TRSV) and the second downhole device is a wireline retrievable safety valve (WLRSV).

8. The downhole tool as recited in claim 7, wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) before or after the wireline retrievable safety valve (WLRSV) is insert within a wellbore.

9. The downhole tool as recited in claim 7, wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) as the wireline retrievable safety valve (WLRSV) is being insert within a wellbore.

10. The downhole tool as recited in claim 3, wherein the first frequency filter is one of a pair of first frequency filters surrounding the first downhole device, and the second frequency filter is one of a pair of second frequency filters surrounding the second downhole device.

11. The downhole tool as recited in claim 1, wherein the first electrical component of the first downhole device is a first electromagnetic assembly and the second electrical component of the second downhole device is a second electromagnetic assembly.

12. A well system, comprising: a wellbore extending through one or more subterranean formations; production tubing disposed in the wellbore; a downhole tool disposed in the wellbore, the downhole tool including: a first downhole device; and a switch system electrically coupled with the first downhole device, the switch system including: an input coupleable to a power source via an electric control line; an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source.

13. The well system as recited in claim 12, wherein the output is a first output coupled to the first electrical component of the first downhole device and a second output coupled to the second electrical component of the second downhole device, and further wherein the first output is coupled to the first electrical component of the first downhole device via the frequency filter or the second output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to switch power between the electric control line and the first downhole device or the electric control line and the second downhole device based upon switching a signal of the power source.

14. The well system as recited in claim 13, wherein the frequency filter is a first frequency filter, and further wherein the first output is coupled to the first electrical component of the first downhole device via the first frequency filter and the second output is coupleable to the second electrical component of the second downhole device via a second frequency filter, the first and second frequency filters configured to switch power between the electric control line and the first downhole device and the electric control line and the second downhole device based upon switching a frequency of the power source.

15. The well system as recited in claim 14, wherein the first frequency filter is a low frequency filter configured to pass a low frequency signal of the power source and block a high frequency signal of the power source, and the second frequency filter is a high frequency filter configured to pass the high frequency signal of the power source and block the low frequency signal of the power source.

16. The well system as recited in claim 14, wherein the first frequency filter is a high frequency filter configured to pass a high frequency signal of the power source and block a low frequency signal of the power source, and the second frequency filter is a low frequency filter configured to pass the low frequency signal of the power source and block the high frequency signal of the power source.

17. The well system as recited in claim 14, wherein the first downhole device further includes a first valve closure mechanism coupled to the first outer housing within the first central bore, and a first bore flow management actuator disposed in the first central bore, the first bore flow management actuator configured to slide from a first initial state to a first subsequent state to move the first valve closure mechanism between a first closed state and a first open state.

18. The well system as recited in claim 17, wherein the first downhole device is a tubing retrievable safety valve (TRSV) and the second downhole device is a wireline retrievable safety valve (WLRSV).

19. The well system as recited in claim 18, wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) before or after the wireline retrievable safety valve (WLRSV) is insert within a wellbore.

20. The well system as recited in claim 18, wherein the switch system is configured to switch power between the electric control line and the tubing retrievable safety valve (TRSV) and the electric control line and the wireline retrievable safety valve (WLRSV) as the wireline retrievable safety valve (WLRSV) is being insert within a wellbore.

21. The well system as recited in claim 14, wherein the first frequency filter is one of a pair of first frequency filters surrounding the first downhole device, and the second frequency filter is one of a pair of second frequency filters surrounding the second downhole device.

22. The well system as recited in claim 12, wherein the first electrical component of the first downhole device is a first electromagnetic assembly and the second electrical component of the second downhole device is a second electromagnetic assembly.

23. A method, comprising: positioning a first downhole device in a wellbore; positioning a second downhole device in the wellbore, wherein a switch system is coupled with the first and second downhole devices, the switch system including: an input coupleable to a power source via an electric control line; an output coupled to a first electrical component of the first downhole device and coupleable to a second electrical component of a second downhole device; and a frequency filter, wherein the output is coupled to the first electrical component of the first downhole device via the frequency filter or the output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to filter power to one of the first downhole device or the second downhole device upon switching a signal of the power source; and switching a signal of the power source to change an operation of the first downhole device or second downhole device.

24. The method as recited in claim 23, wherein the output is a first output coupled to the first electrical component of the first downhole device and a second output coupled to the second electrical component of the second downhole device, and further wherein the first output is coupled to the first electrical component of the first downhole device via the frequency filter or the second output is coupleable to the second electrical component of the second downhole device via the frequency filter, the frequency filter configured to switch power between the electric control line and the first downhole device or the electric control line and the second downhole device based upon switching a signal of the power source.

25. The method as recited in claim 24, wherein the frequency filter is a first frequency filter, and further wherein the first output is coupled to the first electrical component of the first downhole device via the first frequency filter and the second output is coupleable to the second electrical component of the second downhole device via a second frequency filter, the first and second frequency filters configured to switch power between the electric control line and the first downhole device and the electric control line and the second downhole device based upon switching a frequency of the power source.

26. The method as recited in claim 23, wherein the first downhole device includes a first outer housing including a first central bore extending axially through the first outer housing, the first central bore operable to convey subsurface production fluids there through, and the second downhole device includes a second outer housing including a second central bore extending axially through the second outer housing, the second central bore operable to convey subsurface production fluids there through.